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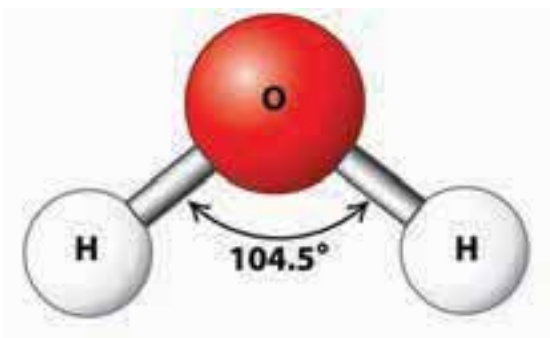


Figure 4.3: Application of rocket fuel

## 4.5 Compounds of Hydrogen

### 4.5.1 Water

Water is one of the most abundant compounds of hydrogen and our earth's surface contains approximately 70 % of ocean which is the major source of water. However, sea water contains many dissolved salts hence it can not be used directly. Water is essential for all living things and our body contains about 65% water.

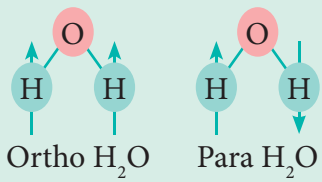






**Ortho-$H_2O$ and Para-$H_2O$**

Water exists in the interstellar clouds, in proto-planetary disks, in the comets and icy satellites of the solar system. In particular, the ortho-to-para ratio (OPR) of water in space has recently received attention. Like hydrogen, water can also be classified into ortho-$H_2O$, in which the spin directions of the nuclei of the hydrogen atoms are parallel, and para-$H_2O$, in which the directions are antiparallel. At the temperature conditions of the earth (300 K), the OPR of $H_2O$ is 3. However, at low temperatures (< 50 K) the amount of para-$H_2O$ increases. It is known that the OPR of water in interstellar clouds and comets has more para-$H_2O$ (OPR = 2.5) than on Earth.




### 4.5.2 Physical Properties:

Water is a colourless and volatile liquid. The peculiar properties of water in the condensed phases are due to the presence of inter molecular hydrogen bonding between water molecules. Hydrogen bonding is responsible for the high melting and boiling points of water. Some of the physical parameters of water are listed in Table 4.2.